	GLOBAL ECONOMICS GRADE: 10 LEARNING EVIDENCE 2 Standard Deviation and Mean (C4) TEACHER'S NAME: Nicolás López Cuéllar	TERM 3 2025–2026

Learning objective: Interpret expected value, variance, and standard deviation from probability distributions by linking graph shape, center, and spread.	**Assessment criteria:** **C4.** Interprets the concept of expected value, variance, and standard deviation of a probability function.

1 Matching Problems

Problem 1. (1) Uniform: match graphs with their means

Correct matches:

Graph	Answer
A	-1
B	2
C	5
D	9

Problem 2. (1) Uniform: match graphs with their spread

Correct matches:

Graph	Answer
A	0.58
B	1.15
C	1.73
D	2.89

Problem 3. (1) Uniform: match graph, mean, and spread

Correct matches:

Graph	(μ, σ)
A	$(2, 1.15)$
B	$(5, 1.73)$
C	$(8, 1.15)$

Problem 4. (2) Uniform: match graph, mean, and spread

Correct matches:

Graph	(μ, σ)
A	$(-1, 1.15)$
B	$(2, 1.73)$
C	$(6, 2.31)$
D	$(9, 1.15)$

Problem 7. (1) Triangular: same spread, different mean

Correct matches:

Graph	Answer
A	$-\frac{4}{3}$
B	$\frac{7}{3}$
C	$\frac{14}{3}$
D	$\frac{28}{3}$

Problem 8. (1) Triangular: same mean, different spread

Correct matches:

Graph	Answer
A	0.41
B	1.08
C	1.47
D	2.04

Problem 9. (1) Triangular: match graph, mean, and spread

Correct matches:

Graph	(μ, σ)
A	$(0, 0.82)$
B	$(4, 1.22)$
C	$(8, 0.41)$

Problem 10. (2) Triangular: match graph, mean, and spread

Correct matches:

Graph	(μ, σ)
A	$\left(-\frac{4}{3}, 0.85\right)$
B	$\left(\frac{7}{3}, 1.25\right)$
C	$\left(\frac{17}{3}, 1.65\right)$
D	$\left(\frac{28}{3}, 0.85\right)$

Problem 13. (1) Normal I: match mean and standard deviation

Correct matches:

Graph	(μ, σ)
A	$(3.2, 0.9)$
B	$(3.9, 1.2)$

Problem 14. (1) Normal II: match mean and standard deviation

Correct matches:

Graph	(μ, σ)
A	$(2.6, 1.1)$
B	$(3.3, 0.9)$
C	$(4.0, 1.3)$

Problem 15. (1) Normal III: match mean and standard deviation

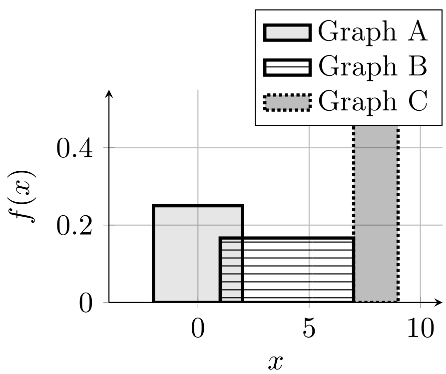
Correct matches:

Graph	(μ, σ)
A	$(2.0, 1.4)$
B	$(3.1, 0.9)$
C	$(4.7, 2.0)$
D	$(5.8, 1.1)$

2 Drawing Problems

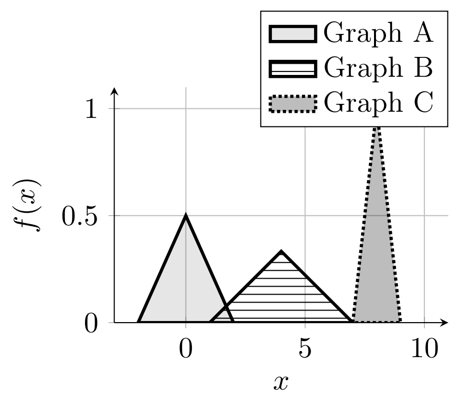
Problem 5. (2) Uniform: draw three uniform distributions

$$A : U[-2, 2], \qquad B : U[1, 7], \qquad C : U[7, 9].$$



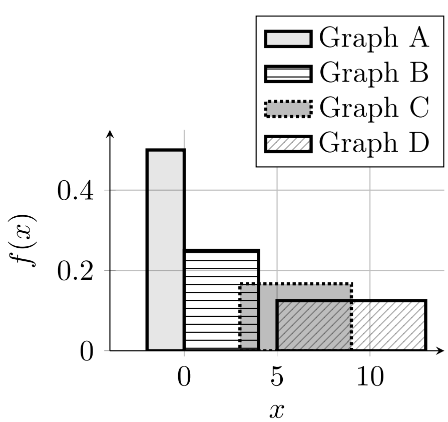
Problem 11. (2) Triangular: draw three triangular distributions

$$A : (a, c, b) = (-2, 0, 2), \quad B : (1, 4, 7), \quad C : (7, 8, 9).$$



Problem 6. (2) Uniform: draw four uniform distributions

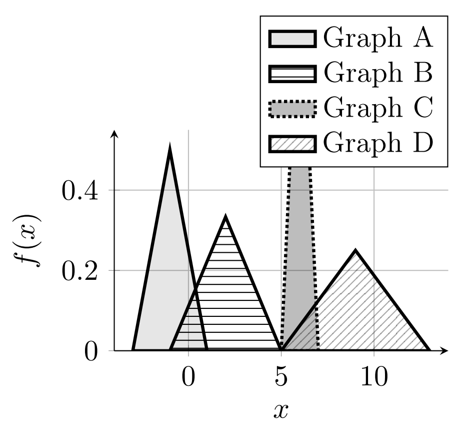
$$A : U[-2, 0], \quad B : U[0, 4], \quad C : U[3, 9], \quad D : U[5, 13].$$



Problem 12. (2) Triangular: draw four triangular distributions

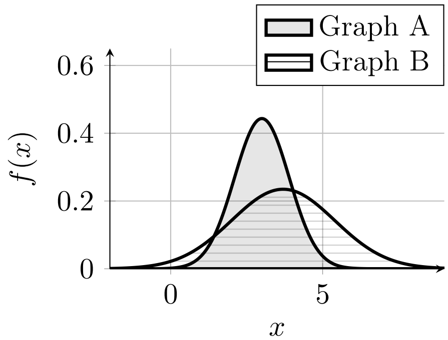
$$A : (a, c, b) = (-3, -1, 1), \ B : (-1, 2, 5)$$

$$C : (5, 6, 7), \ D : (5, 9, 13).$$



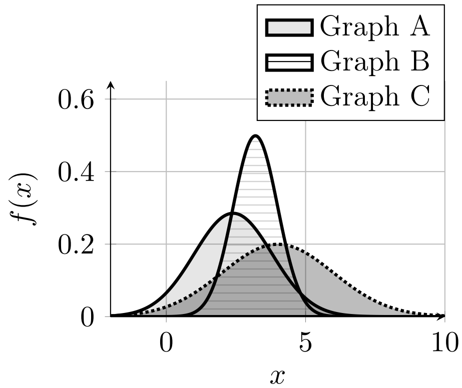
Problem 16. (2) Normal IV: draw two normal distributions

$$A : N(3.0, 0.9^2), \qquad B : N(3.7, 1.7^2).$$



Problem 17. (2) Normal V: draw three normal distributions

$$A : N(2.4, 1.4^2), \quad B : N(3.2, 0.8^2), \quad C : N(4.0, 2.0^2).$$



Problem 18. (3) Normal VI: draw four normal distributions

$$A : N(1.0, 1.2^2), \quad B : N(3.0, 0.7^2)$$

$$C : N(5.0, 1.6^2), \quad D : N(7.0, 1.0^2).$$